

UNB FSK Measurement Field Manual

Pinning down the unknowns in the PortaPack UNB FSK TX app

Read section 1 before connecting anything

A 4 W basestation into an unattenuated receiver front end will destroy it. Section 1.3 is a level budget, not a formality. Section 1.4 validates your measurement chain against a known signal - if you skip it, you will not be able to tell a surprising result from a broken setup, and you will have no way to ask.

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`firmware/application/external/telensa_tx/`

Scope

The physical layer targeted here is publicly confirmed. The protocol is not. Frame structure, deviation, hop generation and the Asia-variant band plan are proprietary and unpublished; the 920-925 MHz plan is inferred from Hong Kong regulation, not from a vendor statement. This manual measures a radio link. It does not describe, and the accompanying app cannot produce, valid PLANet protocol frames.

0. How to use this manual

Every measurement below converts one editable placeholder in the firmware into a measured constant. The procedures are ordered by dependency: M1 establishes where to look, M2 to M4 characterise the physical layer, M5 to M7 open up the frame, and M8 addresses hopping. Work them in order - later measurements assume the earlier results.

Each procedure has the same shape:

- **Pins down** - the firmware value this measurement fixes, with its file and line.
- **Why it matters** - so you can judge whether a shortcut is safe.
- **Setup and Procedure** - the handholding part.
- **Expected result** - what a correct outcome looks like.
- **Acceptance criteria** - the gate a value must pass before it is baked in.
- **If it goes wrong** - the failure modes worth recognising.

The acceptance criteria are the important part.

A single clean burst is never sufficient evidence to bake in a constant. A systematic error - an unlocked reference, an asymmetric filter, a mis-set centre frequency - reproduces perfectly across every burst in a session. Consistency within one capture proves only that the error is stable. That is why the criteria call for multiple devices, multiple sessions, and in one case a deliberate sweep of an analysis parameter to prove the measurement is not an artefact of the tooling.

The bake-in rule

A value graduates from editable field to compiled constant only when it has met every acceptance criterion listed for its measurement. Until then it stays editable, even if you are confident. Section 5 records the gate for each value and where to make the edit.

1. Pre-flight

1.1 Equipment checklist

	Item	Requirement	Why
[]	10 MHz reference	GPSDO or OCXO, ~1 ppb	Mandatory. See 1.2 - this is the most likely thing to invalidate the campaign.
[]	SDR, characterisation	External clock input; 250 ksps+, 16-bit	USRP B210, Airspy R2, or HackRF via CLKIN
[]	SDR, wideband	8-10 Msps, covers 5 MHz in one capture	Stepped sweeps miss a sparse hopper
[]	Band-pass filter	902-928 or 920-925 MHz	Keeps mobile downlink and harmonics out of the front end
[]	Step attenuator	30-40 dB, adequate power rating	Front-end protection. See 1.3
[]	Antennas	Two	Separating uplink from downlink by position and polarisation
[]	PortaPack + UNB FSK TX	App built and installed	Calibration source for 1.4
[]	Toolkit	tools/telensa/, numpy installed	Run 1.4 before you travel, not on arrival
[]	Storage	See 1.6	Wideband captures are large

1.2 Verify the reference is actually locked

Do not take a front-panel LED as proof. Measure it.

1. Set the PortaPack UNB FSK TX app to a known channel and start continuous transmission at low power, well attenuated.
2. Receive it on the characterisation SDR and run the demodulator.
3. Read the reported carrier offset. With both ends locked to the same reference it should be within a few Hz and stable burst to burst.
4. Disconnect the GPSDO and repeat. The offset should visibly worsen. If it does not change at all, your SDR is not using the external reference and you must fix that before proceeding.

Why this matters more than anything else here

The deviation you are trying to measure in M3 may be as small as 125 Hz. At 920 MHz, a 1 ppm reference error is 920 Hz - seven times the quantity of interest. An unlocked receiver does not give you a slightly wrong answer; it gives you a meaningless one, and nothing in the recorded numbers will look obviously wrong afterwards.

1.3 Level budget and front-end protection

A basestation transmits 4 W EIRP - 36 dBm. Inside a screened enclosure there is no free-space loss to save you. Work out the attenuation before connecting, not after.

Quantity	Figure	Note
Basestation TX	+36 dBm EIRP	BS4 manual, US configuration
Telecell TX	+14 to +20 dBm ERP	25-100 mW, datasheet
Typical SDR damage threshold	around 0 dBm	Check your specific hardware; some are lower
Typical SDR compression onset	-20 to -10 dBm	Above this, measurements are distorted even without damage
Target level at the SDR input	-40 to -25 dBm	Comfortable, with headroom for bursts
Attenuation therefore needed	50 to 75 dB from a basestation	Cable and enclosure coupling loss counts towards this

1. Start with maximum attenuation - more than you think you need.

2. Confirm you can see the signal at all, then reduce attenuation in 10 dB steps until it sits in the target window.
3. Check for compression: if reducing attenuation by 10 dB does not raise the reported level by about 10 dB, you are compressed. Back off.
4. Record the final attenuation in the capture log. It is needed to reconstruct absolute levels later.

1.4 Validate the measurement chain

Run this before travelling and again after any change to the setup. It takes two minutes and it is the only thing standing between you and an unresolvable ambiguity on site.

```
cd tools/telesa
./unb_selftest.py --out selftest.cs16
./unb_demod.py selftest.cs16 --expect-rate 500 --expect-deviation 125
```

Expect two PASS lines. Then run the harder cases so you recognise the failure modes when they appear on real data:

```
# Low SNR plus a clock error. Both should PASS.
./unb_selftest.py --out hard.cs16 --snr 10 --freq-offset 900
./unb_demod.py hard.cs16 --symbol-rate-max 1000 \
--expect-rate 500 --expect-deviation 125

# Where deviation breaks down. Expect a FAIL - this is correct.
./unb_selftest.py --out worse.cs16 --snr 6 --freq-offset 900
./unb_demod.py worse.cs16 --symbol-rate-max 1000 --expect-deviation 125
```

If the first test does not pass, stop and fix the toolkit. Do not proceed to real captures with a chain you have not verified.

1.5 Stimulating the system

Traffic is sparse by design: telecells hold their schedules locally and report rarely. Waiting is the slow path, and having the hardware in a cage means you do not have to. In rough order of usefulness:

- **Power-cycle a telecell.** Association and commissioning traffic is the densest and most structured you will see, and the most likely place for addressing to appear in the clear.
- **Occlude the light sensor** to force a dusk or dawn switching transition.
- **Force a fault** - open the lamp circuit, pull the load - to trigger fault reporting.
- **NFC provisioning.** Documented as a feature; what it exchanges is not. Capture the RF side during and just after provisioning.
- **Change exactly one thing at a time, and write it down.** M7 is impossible without this discipline.

Keep a common time reference between your stimulus log and your captures. Correlating "pulled the load at T" with "burst appeared at T plus 2.3 s" is most of the analysis.

1.6 Storage planning

Configuration	Rate	Per hour	Use
10 Msps, cs8	20 MB/s	72 GB	M1, M8 - log only, do not keep IQ
2 Msps, cs8	4 MB/s	14 GB	Wideband, short captures
250 ksps, cs16	1 MB/s	3.6 GB	M2 to M7 - keep this IQ

For the long wideband runs, record detections rather than IQ. A full day at 10 Msps is about 1.7 TB; the same day as a detection log is a few hundred kilobytes. Keep IQ only for bursts you intend to demodulate.

2. What needs pinning down

Every value below is currently an editable field or a documented inference. The right-hand column names the measurement that fixes it.

Value	Current setting	Basis	Fixed by
FSK deviation	125 Hz	Inference from basestation sensitivity. Not published in any form.	M3
Symbol rate	500 bps	FCC test report, but almost certainly rounded	M4
Channel 0 centre, Asia	920.0125 MHz	Inferred from HK regulation, not a vendor statement	M1
Channel count, Asia	200	Derived from the inferred band and raster	M1
Channel raster	25 kHz	FCC test report - confirmed for the US variant, assumed retained	M1
Preamble length	8 bytes of 0x55	Conventional placeholder	M5
Sync word	0x2DD4	Conventional placeholder, no connection to PLANet	M6
Frame layout	None - generic placeholder	Entirely proprietary	M7
Hop generator	16-bit LFSR, seed 0xACE1	Bench substitute, not an attempt to reproduce theirs	M8
Burst length	Derived from frame size	FCC report gives 399.25 ms occupancy	M2

One asymmetry worth planning for

The telecell receive sensitivity is 15 dB worse than the basestation's. That gap is larger than component cost alone explains and hints at a wider downlink receive filter, a higher downlink symbol rate, or a shorter integration window on the node side. If M3 and M4 give different answers for uplink and downlink bursts, that is a real finding, not an error - record both separately rather than averaging them.

3. Measurement procedures

Work these in order. Later procedures depend on earlier results: M2 onward need the channel identified by M1, M5 onward need the bits recovered in M3 and M4, and M7 needs the alignment anchor from M6.

M1. Band, channel raster and channel-0 offset

Pins down	band_defs[] base frequency and channel count (telensa.cpp:33/37/40); UNB_CHANNEL_SPACING (telensa.hpp:67)
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Why it matters

The 920-925 MHz Asia plan is an inference from Hong Kong regulation, not a vendor statement. Nothing else in the manual is safe to trust until you know which band is actually live and where its channels sit.

Setup

- Wideband SDR, 8-10 Msps, GPSDO locked.
- Band-pass filter ahead of the LNA. Attenuate per section 1.3.
- Antenna positioned for the basestation first - it is 16 dB stronger than a telecell and far easier to find.

Procedure

1. Record 60 s centred on 922.5 MHz at 10 Msps. This covers 920-925 in one shot; do not use stepped sweeps, they will miss a sparse hopper.
2. Run the survey against the inferred HK plan.
3. Repeat centred on 915.0 MHz and on 869.5 MHz to rule the other two variants in or out.
4. For whichever band shows energy, note the exact centre frequency of several detections from the CSV.
5. Compute the spacing between adjacent occupied channel centres, and the offset of the lowest one from the round band edge.

```
./unb_survey.py wide_922M5.cs16 --sample-rate 10000000 \
--center 922500000 --band hk --csv m1_hk.csv --hop-report
```

Expected result

Detections roughly 400 ms long on a 25 kHz grid. If the raster is 25 kHz and channel 0 sits 12.5 kHz inside the band edge, the firmware defaults are correct as they stand.

Acceptance criteria - all must hold before baking in

- At least 20 detections, on at least 6 distinct channels.
- Adjacent occupied centres differ by a consistent value within +/-500 Hz. That value is the raster.
- The channel-0 offset is consistent across at least 3 separate captures.

If it goes wrong

- **No energy in any band.** Confirm the system is powered and transmitting - stimulate it per section 1.5 rather than waiting.
- **Energy present but not on a 25 kHz grid.** Do not force the grid. Record the actual centres and re-derive the raster with --band custom.
- **Energy in 920-925 but bursts far shorter than 400 ms.** Likely Sigfox RC4 or UHF RFID, not PLANet. Burst length and hop behaviour are the discriminators, not band occupancy.

M2. Burst envelope, dwell and cadence

Pins down	Confirms the 399.25 ms figure from the FCC test report and establishes the inter-burst interval, which sets the continuous-mode cadence discussed in section 5.
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Why it matters

Burst length times symbol rate bounds the frame size. It is also the cheapest sanity check that you are looking at the right system.

Setup

- Single channel identified by M1. 250 ksps, 16-bit, GPSDO locked.
- Aim for at least 15 dB SNR - see section 1.3.

Procedure

1. Park on the busiest channel from the M1 CSV and record 10 minutes.
2. Run the demodulator and read the per-burst duration column.
3. Record the gap between consecutive burst start times.
4. Repeat on a second channel to confirm the figures are not channel-specific.

```
./unb_demod.py ch_deep.sigmf-data --symbol-rate-max 1000 \
--json-out m2_bursts.json
```

Expected result

Burst durations clustered near 400 ms. Inter-burst gaps will be long and irregular: telecells hold their schedules locally and report sparsely.

Acceptance criteria - all must hold before baking in

- At least 30 bursts measured.
- Duration standard deviation under 5 percent of the mean.
- Mean duration recorded to the nearest millisecond.

If it goes wrong

- **Durations wildly inconsistent.** Your detection threshold is probably clipping burst edges. Lower --threshold and re-run.
- **Almost no bursts in 10 minutes.** Expected. Stimulate the system (section 1.5) instead of extending the wait.

M3. FSK deviation and modulation index

Pins down	UNB_DEFAULT_DEVIATION (telensa.hpp:73)
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Why it matters

This is the single largest gap in the public record and the headline target of the campaign. The FCC report states "FSK 500 bps" and gives no deviation. The current default of 125 Hz is an inference from the basestation sensitivity figure, nothing more.

Setup

- Same single-channel capture as M2 - no new recording needed if that one reached 15 dB SNR.
- GPSDO lock is mandatory here. At 920 MHz a 1 ppm error is 920 Hz, which may be seven times the quantity you are trying to measure.

Procedure

1. Run the demodulator with `--symbol-rate-max` narrowed towards the expected rate. This is the highest-leverage setting for accuracy.
2. Read the deviation figure and the modulation index `h`.
3. Confirm the reported carrier offset is small and stable. A large or drifting offset means your reference is not locked.
4. Re-run with `--symbol-rate-max` at half and at double your first value. The deviation figure should not move materially. If it does, the filter is interacting with the measurement and you should trust the narrowest setting that still reports a reliable symbol rate.
5. Repeat on captures from at least 3 different devices.

```
./unb_demod.py ch_deep.sigmf-data --symbol-rate-max 1000 \
--json-out m3_dev.json
```

Then confirm insensitivity to the filter setting:

```
./unb_demod.py ch_deep.sigmf-data --symbol-rate-max 600
./unb_demod.py ch_deep.sigmf-data --symbol-rate-max 2000
```

Expected result

A deviation between roughly 100 Hz and 2 kHz. Low `h` (around 0.5) is the reading consistent with the published sensitivity; high `h` (around 4) is what Carson's rule gives against "a few kHz occupied bandwidth". Either is plausible - the measurement settles it.

Acceptance criteria - all must hold before baking in

- At least 30 bursts across at least 3 devices.
- Standard deviation under 3 percent of the mean.
- Figure stable within 3 percent across the `--symbol-rate-max` sweep.
- Reported carrier offset under 200 Hz and not drifting between bursts.
- All contributing bursts at 15 dB SNR or better. Below 8 dB the deviation estimate degrades badly - see the accuracy table.

If it goes wrong

- **Deviation changes when you change `--symbol-rate-max`.** The IF filter is clipping the modulation. Widen `--if-bandwidth` explicitly and re-run.
- **Carrier offset of several hundred Hz or more.** Reference not locked, or the wrong channel centre. Fix before recording a number.
- **Deviation varies by more than 10 percent between devices.** Do not average it away. Either the devices genuinely differ, or one capture is bad. Investigate before baking anything in.

M4. Exact symbol rate

Pins down	UNB_DEFAULT_SYMBOL_RATE (telensa.hpp:74)
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Why it matters

The FCC report says 500 bps, which is almost certainly a rounded figure. The exact value reveals the reference oscillator the designers used, and any error here accumulates across a 200-bit frame until bit slicing fails.

Setup

- Same capture as M3. GPSDO locked - a rate measurement inherits your reference error directly.

Procedure

1. Read the symbol rate from the same run as M3.
2. Confirm it is not flagged [UNRELIABLE]. If it is, narrow --symbol-rate-max and --if-bandwidth and re-run.
3. Compare the mean across at least 30 bursts.
4. Check the figure against plausible crystal divisions. A rate of 500.00 bps suggests a synthesised clock; something like 488.28 bps would be a power-of-two division of a 32.768 kHz or 16 MHz reference and is a strong hint about the hardware.

```
python3 -c "import json; d=json.load(open('m3_dev.json')); \
r=[x['symbol_rate_bps'] for x in d if x['rate_reliable']]; \
print(len(r), sum(r)/len(r))"
```

Expected result

A tight cluster. The symbol rate estimator holds to under 0.15 percent even at low SNR once the burst is centred and filtered, so the spread should be very small.

Acceptance criteria - all must hold before baking in

- At least 30 bursts, none flagged [UNRELIABLE].
- Standard deviation under 0.5 percent of the mean.
- Consistent across at least 3 devices and 2 separate sessions.

If it goes wrong

- **Flagged [UNRELIABLE] throughout.** Either SNR is too low, or the signal has no alternating preamble so the spectral cross-check cannot corroborate. Check the rate_fit_quality field in the JSON: above 0.8 the timing measurement is sound even without corroboration.
- **Rate differs between sessions.** Suspect your own reference before suspecting the transmitter.

M5. Preamble pattern and length

Pins down

field_preamble default (ui_telensa_tx.cpp:162) and the 0x55 pattern in build_frame() (telensa.cpp)

Why it matters

The preamble is the receiver's clock-recovery anchor and the first thing you can positively identify in a bitstream. Its length also tells you how much of the ~400 ms burst is overhead.

Setup

- Bits recovered from the M3/M4 capture.

Procedure

1. Export bits with --bits-out.
2. Inspect the leading bits of several bursts by eye. An alternating 1010... run is the preamble.
3. Count its length in bits, and check whether it is the same in every burst.
4. Note the value reported as preamble@N by the demodulator - N is where the alternating run starts, which also tells you your bit alignment.

```
./unb_demod.py ch_deep.sigmf-data --symbol-rate-max 1000 \
--bits-out m5_bits.txt
head -3 m5_bits.txt | cut -c1-96
```

Expected result

A run of alternating bits, most likely 32 to 128 bits, followed by a non-alternating pattern - that transition is the start of the sync word.

Acceptance criteria - all must hold before baking in

- Identical preamble length in at least 20 consecutive bursts.
- Pattern confirmed as strictly alternating, or the actual pattern recorded verbatim if it is not.

If it goes wrong

- **No alternating run.** Possible but unusual. Record the leading bits verbatim and do not force a 0x55 assumption into the firmware.
- **Length varies between bursts.** Check bit alignment first - a one-bit slip makes a constant preamble look variable.

M6. Sync word

Pins down	sym_sync default 0x2DD4 (ui_telensa_tx.cpp:163)
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Why it matters

0x2DD4 is a conventional placeholder borrowed from common radio practice. It has no connection to PLANet. Everything downstream depends on aligning frames correctly, and the sync word is the alignment anchor.

Setup

- Bit file from M5.

Procedure

1. Take the bits immediately following the preamble in one burst.
2. Look for the same run at the same position in every other burst. That common run is the sync word.
3. Determine its length by extending until the bits stop being common across all bursts.
4. Verify by re-running the frame analyser with the candidate. If every burst aligns, it is correct.

```
# Run without --sync first to see raw structure:
./unb_frames.py m5_bits.txt
```

```
# Then confirm the candidate aligns every burst:
./unb_frames.py m5_bits.txt --sync --drop-sync
```

Expected result

The field map will show a constant run after the preamble. "0 burst(s) lacked it" confirms the candidate.

Acceptance criteria - all must hold before baking in

- Candidate found in 100 percent of at least 30 bursts.
- Present in bursts from at least 3 different devices - a sync word is protocol-wide, so if it varies per device it is not a sync word.

If it goes wrong

- **No constant run after the preamble.** The frame may be whitened from the first payload bit. Try M7's whitening check before assuming there is no sync word.
- **Candidate found in only some bursts.** Either two frame types exist, or alignment is slipping. Separate the bursts by length first.

M7. Frame layout: address, payload, CRC

Pins down	Nothing yet - this measurement is what justifies writing a real encoder alongside <code>build_frame()</code> instead of the placeholder.
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Why it matters

This is the measurement that needs deliberate experimental design rather than just a good capture. Addressing and protocol constants are indistinguishable in a single-device capture: both are simply constant. The contrast between devices is the only thing that separates them.

Setup

- **At least 3 devices**, captured back to back with everything else held constant.
- Serial numbers recorded and matched to capture filenames. This is the step people skip and cannot recover later.

Procedure

1. Capture at least 10 bursts from each device separately.
2. Build a combined bit file with one device label per line, in the form label:bits - the tool needs the labels to do the addressing analysis.
3. Run the frame analyser aligned on the M6 sync word.
4. Read the field map. Constant runs are protocol constants. Positions constant within a device but differing between devices are address candidates. Freely varying positions are payload, counter or CRC.
5. Run the CRC search. It requires every frame to validate under the same variant, so a match across 3+ frames is trustworthy.
6. If everything after the sync word is high entropy with no constants at all, suspect whitening - but rule out misalignment first, because misaligned frames also look uniformly random.

```
# Combine per-device bit files with labels:
awk '{print "cellA:"$0}' bits_A.txt > m7_all.txt
awk '{print "cellB:"$0}' bits_B.txt >> m7_all.txt
awk '{print "cellC:"$0}' bits_C.txt >> m7_all.txt
```

```
./unb_frames.py m7_all.txt --sync --drop-sync --crc-search
```

Expected result

A field map with identifiable regions, an address candidate run of 8 to 32 bits, and ideally a CRC match naming a standard variant and the span it covers.

Acceptance criteria - all must hold before baking in

- At least 3 devices, at least 10 bursts each.
- Address candidate run is contiguous and the same width for all devices.
- If a CRC is found, it validates on 100 percent of frames from all devices - not just the ones used to find it.
- Payload regions correlate with something you deliberately changed. If you cannot make a field move by acting on the system, you have not identified it, only located entropy.

If it goes wrong

- **Uniformly high entropy, no constants.** Check alignment, then test common LFSR whitening polynomials. A de-whitened stream shows structure immediately.
- **No CRC match.** It may be a different width, cover a non-byte-aligned span, sit somewhere other than the tail, or the frame may be whitened. Not a failure of the campaign - record and move on.
- **Address candidates scattered rather than contiguous.** Often means the address is bit-interleaved or the frames are misaligned by differing amounts. Re-check per-device alignment individually.

M8. Hop sequence and reproducibility

Pins down	HopSequencer LFSR seed and polynomial (telensa.cpp:78/87, telensa.hpp:117/121)
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Why it matters

15.247 requires a pseudo-random channel list with roughly equal average use. Whether the sequence repeats after a power cycle is the decisive experiment: a repeating sequence means a seeded generator, which is reproducible; a non-repeating one means it is derived from time or node identity, which is not.

Setup

- Wideband capture covering the whole band, as M1.
- Ability to power-cycle a device under controlled conditions.

Procedure

1. Record a long wideband capture - hours, not minutes - and log detections with `--hop-report`.
2. Extract the ordered channel sequence from the CSV.
3. Power-cycle the device. Record again under identical conditions.
4. Compare the two sequences from their respective starts. Identical means a seeded PRNG. Different means time or identity derived.
5. Check that channel usage is roughly uniform across the band, as the regulation requires. A strong bias is informative.
6. Measure the revisit interval per channel and compare against the dwell limits in section 6.

```
./unb_survey.py long_run_1.cs16 --sample-rate 10000000 \
--center 922500000 --csv m8_run1.csv --hop-report
```

After power cycle:

```
./unb_survey.py long_run_2.cs16 --sample-rate 10000000 \
--center 922500000 --csv m8_run2.csv --hop-report
```

```
diff <(cut -d, -f2 m8_run1.csv) <(cut -d, -f2 m8_run2.csv) | head
```

Expected result

A sequence visiting many channels with roughly equal frequency. The two runs will either match from the start or diverge immediately - there is rarely a middle case.

Acceptance criteria - all must hold before baking in

- At least 200 detections per run, covering at least 50 channels.
- Reproducibility result the same across at least 2 power cycles.
- Channel usage histogram recorded, whether uniform or not.

If it goes wrong

- **Too few detections to see a pattern.** Extend the capture or stimulate harder. Hop analysis is the most data-hungry measurement here; budget hours.
- **Sequences match for a while then diverge.** Suggests a seeded generator that is re-seeded periodically or resynchronised to the basestation. Note where divergence begins.

5. Baking in a measured value

When a value has met every acceptance criterion for its measurement, make the edit below. Change the comment at the same time: each placeholder is currently annotated as an inference, and that annotation must become a citation of your measurement - date, session, sample size. A constant with a stale comment claiming it is a guess is nearly as bad as a guess.

Value	Edit here	Also update
FSK deviation	telensa.hpp:73 UNB_DEFAULT_DEVIATION	The comment above it, which currently explains the h=0.5 inference
Symbol rate	telensa.hpp:74 UNB_DEFAULT_SYMBOL_RATE	Nothing else - samples_per_bit derives from it
Asia channel plan	telensa.cpp:33 band_defs[BAND_HK]	The comment marking it INFERRED, and README section 1
Channel raster	telensa.hpp:67 UNB_CHANNEL_SPACING	All three band_defs channel counts depend on this
Preamble	ui_telensa_tx.cpp:162, and build_frame() if the pattern is not 0x55	telensa.hpp scope comment
Sync word	ui_telensa_tx.cpp:163 sym_sync.set_value()	telensa.hpp scope comment
Frame layout	New encoder alongside build_frame() in telensa.cpp	Do not redefine the placeholder - add beside it, per the note in telensa.hpp
Hop generator	telensa.cpp:78/87 and telensa.hpp:117/121	The comment disclaiming any relationship to Telensa's generator

If a measurement contradicts the public record

Record what you measured, and keep the citation to what was published alongside it. A conflict between a vendor filing and your bench is a finding in its own right, and the next person needs both numbers to make sense of it. Do not quietly overwrite one with the other.

Regenerating this manual

The line references above are baked into make_field_manual.py. If you move a constant, update the manual generator and regenerate, so the document does not drift out of step with the code:

```
python3 make_field_manual.py --out UNB_FSK_Field_Manual.pdf
```

Photocopy or reproduce these. The fields that get omitted in the moment are the ones that make a capture unusable later - especially "what was stimulated" and the reference lock state.

[illegible][illegible]

Sweep check	--symbol-rate-max at half: _____ Hz nominal: _____ Hz double: _____ Hz stable within 3%? [] yes [] no
-------------	--

[illegible]

6.4 M5 to M7 frame findings

Finding	Value	Evidence
Preamble pattern		
Preamble length (bits)		
Sync word		
Sync length (bits)		
Address offset (bits)		
Address width (bits)		
CRC variant		
CRC span		
Whitening suspected?		
Frame length (bits)		

6.5 M8 hop findings

Finding	Run 1	Run 2 (after power cycle)
Detections		
Unique channels		
Sequence reproducible?		
Divergence point, if any		
Revisit interval, median		
Usage roughly uniform?		

6.6 Sign-off: values cleared for bake-in

Value	Measured	Criteria met?	By	Date

7. Troubleshooting

7.1 No bursts detected

Work down this list in order. Each step is cheaper than the next.

1. Lower --threshold. A weak burst may be only a few dB above the floor.
2. Lower --min-duration in case bursts are shorter than expected.
3. Check --channel-offset - the signal may not be at DC.
4. Confirm --sample-rate. A wrong rate breaks everything downstream and produces confidently wrong numbers rather than an error.
5. Narrow --symbol-rate-max. This tightens the IF filter and is the single most effective change for a weak slow signal.
6. Widen --afc-range if you suspect a large clock offset.
7. Pass --whole-file if the capture has been pre-trimmed to a single burst and therefore has no idle region to measure a floor against.
8. Verify with the selftest that the chain still works at all.

7.2 Symbol rate flagged UNRELIABLE

- Check rate_fit_quality in the JSON output. Above 0.8, the timing measurement is sound even if the spectral cross-check could not corroborate it - which happens legitimately when there is no alternating preamble.
- Narrow --symbol-rate-max towards the expected rate.
- Confirm SNR. The rate estimate is robust once the burst is centred and filtered, so persistent unreliability usually means the burst is not being centred - check the reported carrier offset.

7.3 Deviation figure looks implausible

- Sweep --symbol-rate-max. If the deviation moves, the IF filter is clipping the modulation - widen --if-bandwidth explicitly.
- Check the carrier offset. A large offset used to compress the deviation badly; AFC now corrects for it, but only within --afc-range. Outside that range the burst is simply filtered away.
- Confirm SNR is at least 15 dB. Below 8 dB the deviation estimate degrades to the point of being useless while still looking plausible.

7.4 Frame analysis shows no structure

- Rule out misalignment first. Without a correct sync anchor every position looks random, which is indistinguishable from whitening.
- Separate bursts by length before analysing - two frame types mixed together will smear every field.
- Check bit alignment per device. Different devices may be offset by different amounts if their captures were trimmed differently.
- Only then consider whitening or encryption, and test common LFSR polynomials before concluding it is crypto.

7.5 Accuracy reference

Measured against synthetic signals at 500 bps, $h = 0.5$, with --symbol-rate-max 1000. SNR is quoted across the full capture bandwidth, which is the pessimistic convention - in-band SNR after the IF filter is considerably better.

Capture SNR	Deviation error	Symbol rate error	Verdict
20 dB or better	under 1%	under 0.1%	Both trustworthy
15 dB	about 1.4%	under 0.1%	Both trustworthy - target this
10 to 12 dB	2 to 3%	under 0.1%	Rate good, deviation marginal
8 dB	about 3.4%	about 0.1%	Rate good, deviation weak

Capture SNR	Deviation error	Symbol rate error	Verdict
6 dB	about 20%	about 0.15%	Rate only. Discard deviation

8. Quick reference

8.1 Commands

```
# Validate the chain (do this first, every session)
./unb_selftest.py --out selftest.cs16
./unb_demod.py selftest.cs16 --expect-rate 500 --expect-deviation 125

# M1 Which band and channels are live
./unb_survey.py wide.cs16 --sample-rate 10000000 --center 922500000 \
--band hk --csv m1.csv --hop-report

# M2-M4 Physical layer on one channel
./unb_demod.py deep.sigmf-data --symbol-rate-max 1000 \
--bits-out bits.txt --json-out bursts.json

# Pull one channel out of a wideband capture
./unb_demod.py wide.cs16 --sample-rate 2000000 --channel-offset -75000

# M5-M7 Frame structure, needs device labels for addressing
./unb_frames.py all.txt --sync --drop-sync --crc-search
```

8.2 Settings that matter most

Setting	Default	When to change it
--symbol-rate-max	4000 bps	Lower towards the true rate. The highest-leverage setting for weak signals - it also sets the auto IF bandwidth
--if-bandwidth	2.5x rate max	Widen if the deviation figure shifts when you sweep symbol-rate-max
--afc-range	+/-3 kHz	Widen for a large clock offset; keep under half the channel spacing so it cannot lock onto the neighbour
--threshold	10 dB	Lower for weak bursts
--min-duration	20 ms	Lower if bursts are shorter than expected
--fft-size (survey)	8192	Raise for finer RBW. Aim for hundreds of Hz, not tens of kHz

8.3 Figures worth remembering

Quantity	Value	Status
Symbol rate	500 bps	Confirmed, FCC test report
Channel raster	25 kHz	Confirmed for US variant
Burst occupancy	399.25 ms in 20 s	Confirmed, measured by test house
US band	910.4875 - 919.9875 MHz	Confirmed
EU band, BS TX	869.40 - 869.65 MHz	Confirmed, BS4 manual
Asia band	920 - 925 MHz	INFERRED from HK regulation
FSK deviation	unknown	Not published. Target of M3
Basestation TX	4 W EIRP (36 dBm)	Confirmed
Telecell TX	25 - 100 mW ERP	Confirmed, datasheet
Basestation sensitivity	-139 dBm conducted	Confirmed
Telecell sensitivity	-124 dBm	Confirmed - note the 15 dB gap
1 ppm at 920 MHz	920 Hz	Why the GPSDO is mandatory

Before you leave the site

Check that every capture in the log has its metadata: centre frequency, sample rate, gain, attenuation, antenna, timestamp, reference lock state, device serial, and what you stimulated. A capture missing the last of those cannot contribute to M7 and cannot be reconstructed afterwards. Confirm the acceptance criteria you believe you have met, while you are still able to take another capture.